Mohammed Batis

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Page: https://batis1.github.io/batis1/
Date of Birth: 09/12/1998 Nationality: Yemeni

Research Interests

I study the mechanistic interpretability of language-only LLMs, combining game-theoretic interactions with causal analysis to understand how model components jointly shape behavior.

I enjoy design engineering, logic, and ideation, with a broad interest in how ideas become thoughtfully designed systems. I am currently reading about AI accelerator hardware, chip limitations, and the design decisions shaped by those constraints.

Education

PhD Computer Science and Technology 2025 – Present
University of Science and Technology of China, Hefei, China
Voluntary Withdrawn

MSc Computer Science and Technology 05/10/2022 – 30/07/2025
Wenzhou University
Final grade: 4.5. Thesis: Research on the Improved RIME Optimization Algorithm and Its Application for Wrapper-based Feature Selection in Biomedical Data.

BSc Computer Science and Technology 07/10/2018 – 14/06/2022
Wenzhou University
Final grade: 4.03 GPA. Thesis: Design and Implementation of a Chinese HSK Test Preparation Platform.

Secondary School 14/08/2014 – 09/07/2017
Mukalla Model Secondary School
Final grade: 98.63%.

Publications

Google Scholar: 24 citations, h-index: 2.
Citation metrics updated October 27, 2026. Journal metrics: 2025 JCR.

Random differential RIME optimization for multi-threshold segmentation in dermoscopic skin cancer images 2026
Biomedical Signal Processing and Control, 123, 110533
M. Batis, Q. Shao, L. Liu, A. A. Heidari, H. Chen.
Impact factor: 5.7; JCR Q2 (2025). 0 citations. Article link.

Prey capture enhanced Harris hawks optimizer for wrapper-based feature selection in high-dimensional medical data 2026
Computer Methods and Programs in Biomedicine, 109237
M. Batis, Y. Chen, L. Liu, A. A. Heidari, H. Chen.
Impact factor: 6.4; JCR Q1 (2025). 3 citations. Article link.

ACGRIME: adaptive chaotic Gaussian RIME optimizer for global optimization and feature selection 2025
Cluster Computing, 28(1), 61
M. Batis, Y. Chen, M. Wang, L. Liu, A. A. Heidari, H. Chen.
Impact factor: 5.5; JCR Q1 (2025). 21 citations. Article link.

Projects

WHSK – Interactive HSK Preparation Platform 09/01/2022 – Current
Independent academic and software project
Developed a gamified web platform for HSK exam preparation, featuring interactive practice and progress tracking. Registered software copyright in China (No. 2025SR0444242). Project link: https://whsk.tech/.

Win TOEFL – Interactive TOEFL Practice Platform 2026 – Present
Independent software project
A comprehensive TOEFL practice platform with reading, listening, speaking, and writing exercises, full-length tests, progress tracking, and vocabulary tools. Project link.

CNN Lens – Interactive CNN Visualization 2026 – Present
Independent software project
An interactive visual learning tool for exploring convolutional neural networks, from image channels and learned filters to feature maps, dense layers, and class predictions. Project link.

Honours and Awards

- CAS-ANSO Scholarship at USTC (2025)
- Chinese Government Scholarship at Wenzhou University (10/07/2022)
- Excellent Graduation Design Award at Wenzhou University (20/06/2022)
- Zhejiang Government Scholarship at Wenzhou University (09/10/2021)
- Second Place in Mukalla Model Secondary School Rankings (07/12/2017)
- 9th Best Middle School Graduate in the Province of Wadi Hadramout (04/07/2013)

Technical Skills

- Programming: Python, JavaScript, MATLAB, PyTorch
- Web and Software: React, Svelte, Node.js, MongoDB, MySQL

Languages

- Arabic (mother tongue)
- English
- Chinese

Hobbies

Going to the gym
In 3 months, I went from 85 to 69 kg and reduced my body fat from 27% to 15%.

Typing
I learned touch typing on TypingClub, spending about 72 hours over a year and a half to complete the course. My current typing speed is about 70 words per minute.